Summary in Graph

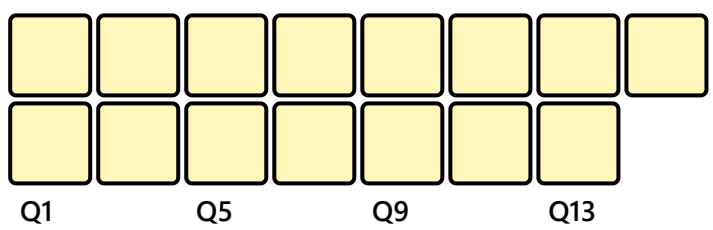
Exam Summary (GO Classes 2026 | Algorithms | Sorting Algorithms | Weekly Quiz 1)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (10 + 5)
Total Marks:	20 (10 + 10)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (36) out of 364 users

EXAM RESPONSE EXAM STATS FEEDBACK

Technical

Q #1 Multiple Select Type Award: 1 Penalty: 0 Algorithms

For Quick-Sort, let $\text{comps}(n)$ be the number of comparisons required to sort n elements, where each partition step performs $n-1$ comparisons.

Consider the following recurrences:

1. $\text{comps}(n) = 2 \cdot \text{comps}(n/2) + (n-1)$

2. $\text{comps}(n) = \text{comps}(n-1) + (n-1)$

Which of the following statements are true?

A. Recurrence (1) leads to $O(n \log n)$ comparisons, and this happens when the pivot always splits the array into two equal halves.

B. Recurrence (2) leads to $O\left(n^2\right)$ comparisons, and this happens when the pivot is always the smallest or largest element.

C. Recurrence (1) arises when the array is already sorted.

D. Recurrence (2) arises when the pivot is always the median of the current array.

Your Answer: | Correct Answer: A;B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the array $0, 4, 2, 7, 6, 1, 3, 5$ being sorted using Insertion Sort.

What will be the array after inserting the element 3 into its correct position?

A. 01234675

B. 01243675

C. 01246735

D. 01234567

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Merge Sort is applied on the array $0, 4, 2, 7, 6, 1, 3, 5$.

During its execution, several intermediate arrays are produced.

Which of the following arrays cannot appear as an intermediate step in any correct execution of Merge Sort?

A. 02471635

B. 01243567

C. 04271356

D. 01234567

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Which sorting algorithm most likely produced the shown execution trace?

Initial array: $0, 4, 2, 7, 6, 1, 3, 5$

During execution: $0, 2, 4, 7, 6, 1, 3, 5$

Final sorted result: $0, 1, 2, 3, 4, 5, 6, 7$

A. In-place Heapsort

B. Selection Sort

C. Insertion Sort

D. Merge Sort

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Consider the number of element swaps performed by selection sort and bubble sort in the worst case for a list of size N.

Which of the following statements are correct?

A. Selection sort always performs exactly ($N - 1$) element swaps.

B. The number of element swaps in selection sort depends on the initial arrangement of elements.

C. Bubble sort, in the worst case, performs $\frac{(N-2)(N-1)}{2}$ element swaps.

D. Bubble sort performs the same number of element swaps for both sorted and reverse-sorted arrays.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following array:

$$12, 18, 27, 35, 46, 58, 72, 85$$

Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

A. Selection sort

B. Mergesort

C. Insertion sort

D. Quicksort using the last element as pivot

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

You are given an array A of n distinct integers in sorted order. The task is to determine whether there exists an index i such that $A[i] = i$.

Which of the following statements are correct?

A. A naive algorithm that checks every index runs in $O(n)$ time.

B. If all integers are non-negative ($A[i] \geq 0$), the problem can be solved in $O(1)$ time.

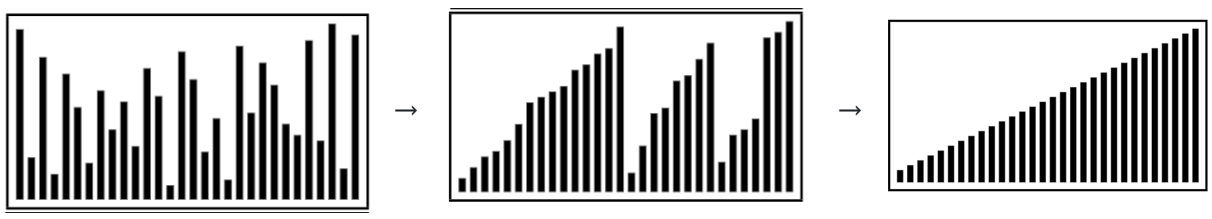
C. If all integers are positive, then no valid index can exist.

D. The problem always requires $O(n)$ time, regardless of the input conditions.

Your Answer: | Correct Answer: A;B;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following diagrams showing intermediate steps while sorting a collection of sticks by height:



Which sorting algorithm might produce such intermediate steps?

A. Selection Sort

B. Quicksort

C. Insertion Sort

D. Merge Sort

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Professor Fiorina uses the following algorithm for merging k sorted lists, each having $\frac{n}{k}$ elements:

- She merges the first two lists using a linear-time merge procedure.
- Then she merges the resulting list with the third list, then with the fourth, and so on, until all lists are merged into a single sorted list of n elements.

What is the worst-case running time of this algorithm in terms of n and k?

A. $\Theta(n \log k)$

B. $\Theta(n + k)$

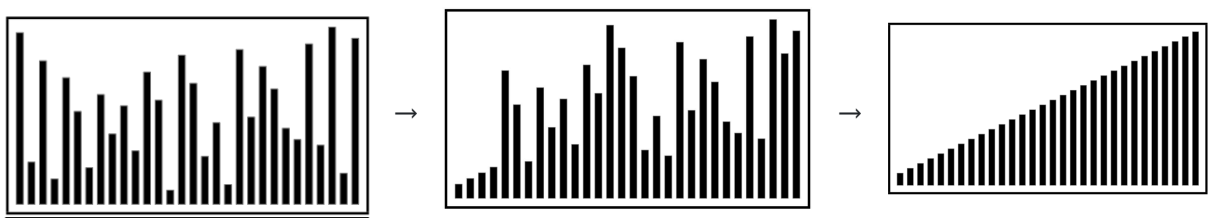
C. $\Theta(nk)$

D. $\Theta\left(\frac{n}{k}\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the following diagrams showing intermediate steps while sorting a collection of sticks by height:



Which sorting algorithm might produce such intermediate steps?

A. Selection Sort

B. Quicksort

C. Insertion Sort

D. Merge Sort

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following algorithm:

```
Process(A):

    done = false
    while done == false:
        done = true
        for i = 0 to n-3:
            if A[i] > A[i+2]:
                reverse elements from i to i+2
                done = false
```

Which of the following is correct?

A. The algorithm always sorts the array correctly for all possible inputs

B. The algorithm may fail to sort the array for some inputs

C. The algorithm never terminates for certain inputs

D. The algorithm works correctly only if the array size is even

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 01sec | Discuss

Q #12 Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Assume a variant of **Quicksort** in which, at every recursive step, the input array of size n is deterministically partitioned into two subproblems:

where $0 < \alpha \leq \frac{1}{2}$.

What is the asymptotic expression for the **maximum depth** of the recursion tree in terms of n and α?

A. $-\dfrac{\lg n}{\lg(1-\alpha)}$

B. $\dfrac{\lg n}{\lg(1-\alpha)}$

C. $-\dfrac{\lg n}{\lg \alpha}$

D. $\dfrac{\lg n}{\lg \alpha}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Imagine $\mathbf{n}$ people standing in a straight line, each holding a card with a number on it.
They follow the following method to sort the cards:

- In the first round, every even-positioned person (2nd, 4th, 6th, ...) compares with their right neighbor and keeps the smaller number on the left, larger on the right.
- In the next round, every odd-positioned person (1st, 3rd, 5th, ...) does the same.
- Rounds alternate until the line is fully sorted.

In the worst case, how many rounds are required?

A. $n \log n$

B. n^2

C. n

D. $n^{1.5}$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14

Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Randomized QuickSort (where the pivot is chosen uniformly at random from the current subarray) is applied on an array of 10 distinct numbers. Suppose we focus on the elements 25 and 70, which in the sorted order look like this:

$$10, 18, 22, 25, 31, 40, 53, 66, 70, 85$$

What is the probability that 25 and 70 are directly compared at some stage of QuickSort?

A. $1/10$

B. $2/7$

C. $1/3$

D. $1/5$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider the problem of sorting an array of size n using only the following two operations:

$\text{compare}(\mathbf{x}, \mathbf{y})$: compares two elements x and y.

flip(i): reverses the order of the first i elements of the array.

For example, if the array is $[1, 4, 6, 5, 3, 10, 1]$, then applying flip(4) results in $[5, 6, 4, 1, 3, 10, 1]$.

The Pancake Sort algorithm works as follows:

In each iteration,

- find the maximum element in the current prefix of size k,
- flip the prefix to bring this maximum element to the front (if needed),
- flip the first k elements to move the maximum into its correct final position at index k.

This process is repeated for $k = n, n-1, \ldots, 2$.

Which of the following statements is/are correct about Pancake Sort?

A. After the i-th iteration, the last i elements of the array are in their correct sorted positions.

B. In each iteration, the algorithm always requires exactly 1 flip in the worst case.

C. In the worst case, the total number of flips performed by the algorithm is at most $2n$.

D. The number of comparisons required by the algorithm is $O\left(n^2\right)$.

Your Answer: | Correct Answer: A;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

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